Linear and exponential functions

1 Exponential

2

a

n	1	2	3	4	5
$f(n)$	6	11	16	21	26
$g(n)$	6	12	24	48	96

b The consecutive terms of $f(n)$ are increasing linearly by 5, while the consecutive terms of $g(n)$ are increasing by a factor of 2.

c John is correct, since the second term of $g(n)$ is larger than $f(n)$, and the terms of $g(n)$ increase at a faster rate they will remain larger than the terms of $f(n)$.

3 Only for very large values of x.

4 Neville is incorrect. $g(x) > f(x)$ for some values of $x \geq 1$, but $f(x)$ is increasing more rapidly, so at some point $f(x)$ will become greater in value.

5 Ivan is incorrect. $g(x) > f(x)$ for some values of $x \geq 1$, but $f(x)$ is increasing more rapidly so at some point $f(x)$ will become greater in value.

6

a 4 units

b 2

c The exponential function

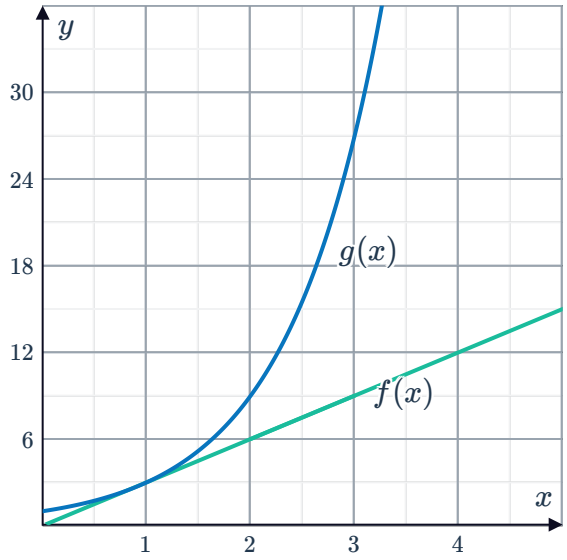
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7

a

x	0	1	2	3	4
$f(x)$	0	3	6	9	12
$g(x)$	1	3	9	27	81

b



c

i 3 **ii** 3

d $g(x) = 3^x$

e

i 24 **ii** 6561

f $f(x) = 3x$ increases at a constant rate. $g(x) = 3^x$ increases at an increasing rate.

8

a

x	4	5	6
$f(x)$	22	27	32
$g(x)$	405	1215	3645

b

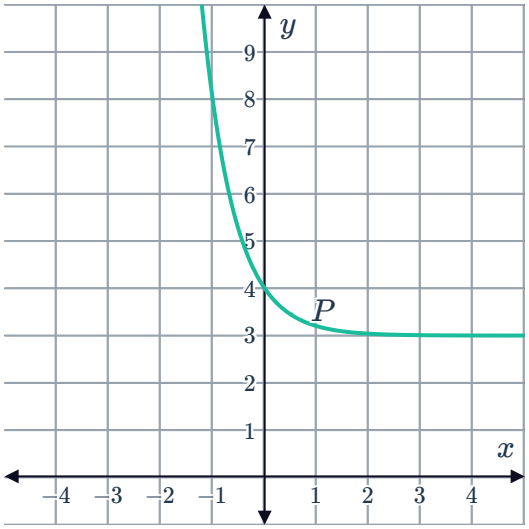
i 5 **ii** 3 **iii** 5 **iv** 3

c The linear function increases by a constant difference and the exponential function increases by a constant multiplicative factor.

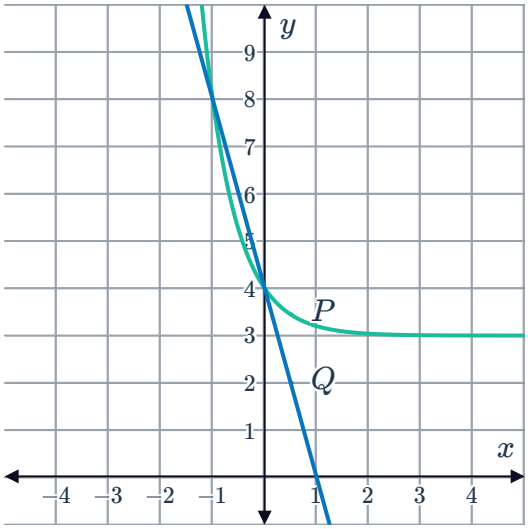
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9

a



b

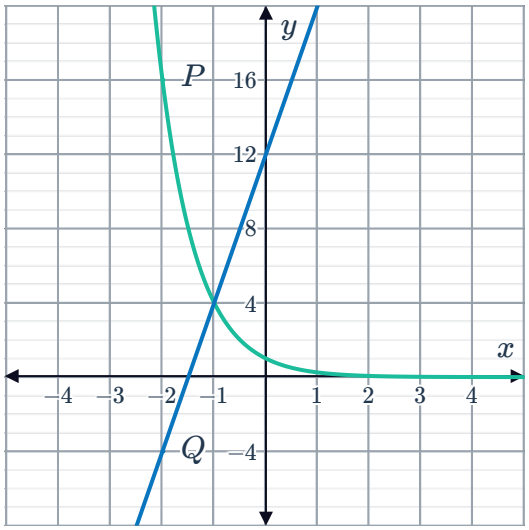


c Twice

d P

10

a



b Once

c Q

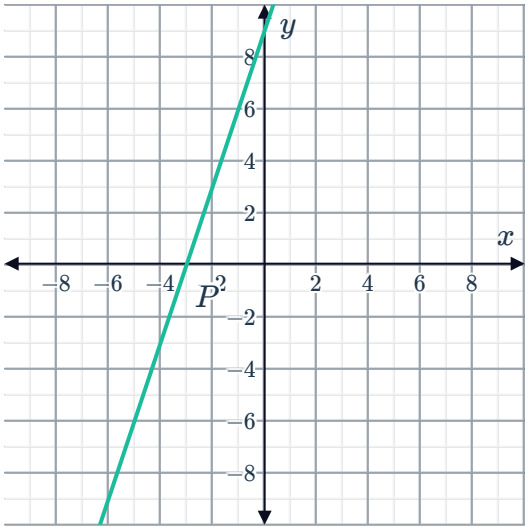
d Q

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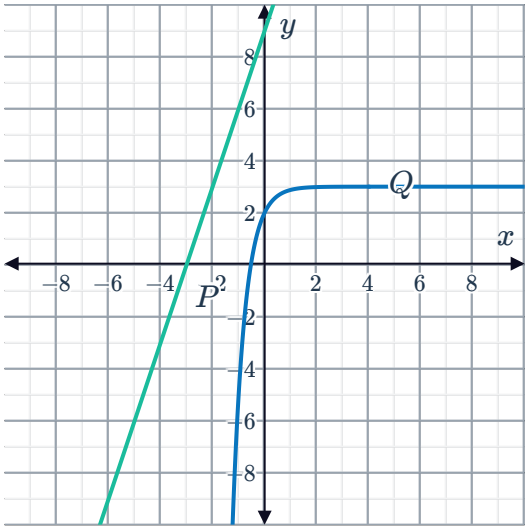
11

a Function P is linear and function Q is non-linear.

b



c



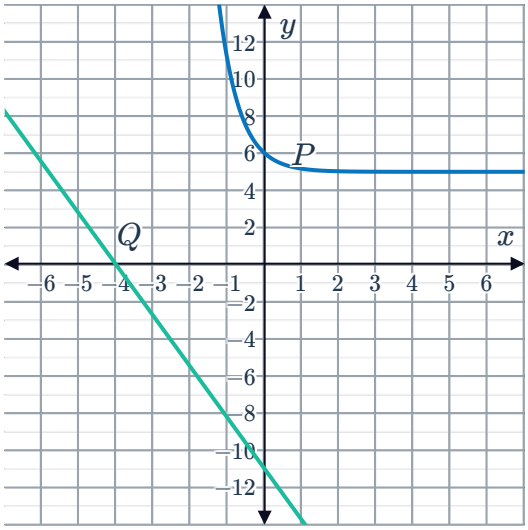
d 0

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a P is an exponential function and Q is a linear function.

b



c P

d Zero times

Applications

13

a Treatment A **b** 5 **c** Treatment A **d** 72 550

14

a $f(n) = 125\,000 + 3000n$

b $g(n) = 20\,000\,(1.4)^n$

c

n	2	3	4
Increase in salary compared to previous year when working for a company	3000	3000	3000
Increase in salary compared to previous year when working for her own business	11 200	15 680	21 952

d Working for a company

e Starting her own business

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15

a

x	0	3	6	9
Company A's bill	200	218	236	254
Company B's bill	200	224.97	253.06	284.66

b Company B

16 Sophia

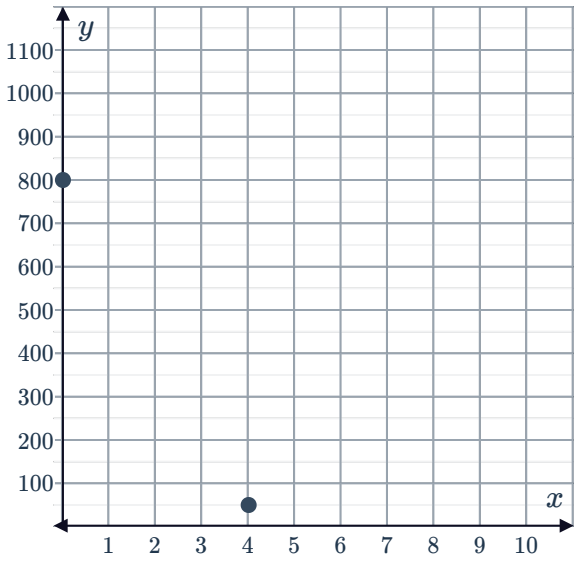
17

a Adravil, because the bacteria are only growing linearly, increasing at 12 per hour, while with Felicium they are growing exponentially, tripling every hour, which tends to infinity much quicker.

b In the short-term, Tretonin is better than Adravil in the first 24 hours because it is growing by less than 12 microbes per hour, but the rate of growth is increase by 6 every three hours. In the long-term, Adravil is better than Tretonin because after 24 hours it will be increasing at a faster rate than Adravil as it is quadratic growth versus linear.

18

a

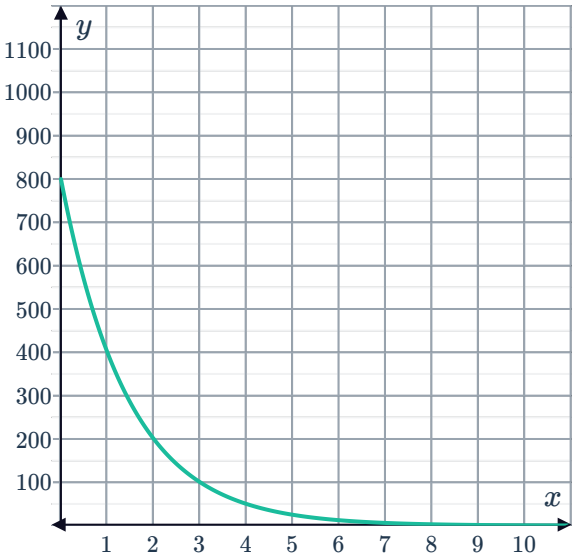


b Exponentially

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c $V = 800\,(0.5^t)$

d



e Less by $46.88

19

a

x	0	4	8	12
Massive Dynamic's bill	200	268	336	404
Stark Industry's bill	200	252.50	318.77	402.44

b A: Massive Dynamic, B: Stark Industry

c Stark Industry

d $21 159

e Example answers:

- Quit Stark Industry and sign up for Massive Dynamic instead.
- Pay his bills on time.

20

a

n	1	2	3	4
Income in year n with Kord Enterprise	125 000	174 000	223 000	272 000
Income in year n with her own business	40 000	64 000	102 400	163 840

b A: Working for Kord Industry, B: Working for herself

c Working for Kord Enterprise.

d Working for her own business.

e No

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